Goal-Conditioned Reinforcement Learning for Atari Seaquest

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Abstract

Goal-conditioned reinforcement learning (GCRL) enables agents to pursue multiple objectives within a single framework. We investigate the application of GCRL with Hindsight Experience Replay (HER) to Atari Seaquest, a game where vanilla DQN achieves only 6% of human performance. Seaquest requires coordinating multiple objectives – rescuing divers, managing oxygen, and avoiding enemies – making it a natural candidate for goal-conditioned approaches.

We implement a goal-conditioned Double DQN with HER, representing goals as two-dimensional vectors [divers collected, resurfaces]. Initial experiments with sparse binary rewards proved ineffective, leading us to develop a dense reward shaping approach while maintaining HER's goal-relabeling mechanism. We also employ two-phase curriculum learning to progressively increase goal difficulty and mixed priority/uniform replay to balance learning across curriculum levels.

Training DQN+HER for 10 million steps yielded mixed results: similar average performance to baseline DQN (≈200-250 points) but occasional high-reward episodes (>1000 points). The agent reliably learned to collect up to 4 divers but plateaued at this curriculum level. Replacing DQN with Soft Actor-Critic (SAC) substantially improved performance, reaching 6-diver collection within 20 million steps and demonstrating steadily increasing rewards.

Our results show that HER enables goal-directed learning in Seaquest, but exploration strategy significantly impacts effectiveness: SAC's entropy-regularized exploration outperformed DQN's epsilon-greedy approach. This contrast between epsilon-greedy (DQN) and entropy-regularized (SAC) exploration highlights challenges in applying GCRL techniques beyond their original sparse-reward robotics domains.

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1. Introduction

Many problems in reinforcement learning (RL) are framed within the context of a single objective. However, multi-objective decision making is a common problem in many real-world scenarios.

Goal-conditioned reinforcement learning (GCRL) is a variant of reinforcement learning that allows the flexible specification of objectives. In GCRL, the agent is trained to learn a *goal-conditioned policy* $\pi(a \mid s, g)$ that informs an action a based on not only the current state s but also the goal g on which it is conditioned (Liu et al., 2022). This allows the agent to act within the context of multiple objectives.

In this work, we investigate how GCRL can be applied to Atari's *Seaquest* video game. Seaquest is a single-player game in which the player controls a submarine and aims to maximize their score by shooting enemies and rescuing divers, all while periodically resurfacing for oxygen (Figure 1). Seaquest presents a particularly intriguing opportunity for GCRL because these competing objectives form natural goal candidates.

Moreover, Seaquest has historically been a challenging benchmark for RL. While Mnih et al. (2013) trained an RL agent to play Seaquest using a deep Q-network (DQN) to maximize change in reward, the agent performed much more poorly compared to humans (6% of human performance). This performance can be attributed to the game's emphasis on long-term strategies and coordination of multiple objectives.

Finally, while GCRL has shown promise in other video games – such as universal value function approximators (UVFAs) on Ms. Pac-Man (Schaul et al., 2015) – Seaquest remains unexplored in this context.

We apply Hindsight Experience Replay (HER) – a common GCRL technique (Andrychowicz et al., 2017) – to train a DQN agent playing Seaquest. Besides HER, we also expand on the classical DQN algorithm by introducing curriculum learning and reward shaping. Our experiments reveal both promise and challenges in applying GCRL to Seaquest. While our HER-based approach enables goal-

Figure 1. In Seaquest, the player controls a submarine (A) and aims to maximize their score (B) by shooting enemies (C) and rescuing divers (D). They must periodically resurface to refill their oxygen (E) and to unload current divers on the submarine (F). The player has a limited number of lives (G), which are lost if the submarine makes contact with an enemy or resurfaces without a diver aboard. (Source: YouTube)

directed learning and occasional high-reward episodes, it does not consistently outperform baseline DQN in average performance.

2. Background and Related Work

2.1. Deep Q-Networks

Deep Q-Networks (DQN) (Mnih et al., 2013) revolutionized reinforcement learning by successfully combining Q-learning with deep neural networks to learn policies directly from high-dimensional sensory input. DQN learns an action-value function (or Q-function) $Q(s, a)$ that estimates the expected cumulative reward from taking action a in state s and following the optimal policy thereafter. The key innovations enabling stable learning include *experience replay*, which stores transitions in a replay buffer for batch sampling, and a separate target network Q_{target} that is periodically updated to stabilize training.

The DQN objective minimizes the *temporal difference* (TD) error:

$$\mathcal{L}(\theta) = \mathbb{E}_{(s,a,r,s')\sim\mathcal{D}}\Big[\big(r + \gamma \max_{a'} Q_{\text{target}}(s', a'; \theta^-) - Q(s, a; \theta)\big)^2\Big] \tag{1}$$

where $\mathcal{D}$ is the replay buffer, γ is the discount factor, and θ^- denotes the target network parameters.

Double DQN (van Hasselt et al., 2015) addresses overestimation bias in standard DQN by decoupling action selection from action evaluation:

$$y = r + \gamma Q_{\text{target}}(s', \arg\max_{a'} Q(s', a'; \theta); \theta^-) \tag{2}$$

This modification improves learning stability and is widely adopted in modern DQN implementations.

The Arcade Learning Environment (ALE) (Bellemare et al., 2013) provides a standardized platform for evaluating RL algorithms on Atari 2600 games. Seaquest, one of the ALE benchmarks, proved particularly challenging for early DQN agents, achieving only 6% of human-level performance (Mnih et al., 2013) compared to superhuman performance on simpler games like Pong or Breakout.

2.2. Goal-Conditioned Reinforcement Learning

Goal-conditioned reinforcement learning (GCRL) extends the standard RL framework to handle multiple objectives by conditioning policies and value functions on explicit goals. Rather than learning a single policy $\pi(a|s)$, GCRL learns a family of policies $\pi(a|s, g)$, where g represents the desired goal (Liu et al., 2022).

Universal Value Function Approximators (UVFAs) (Schaul et al., 2015) formalize this approach by learning a single function $V(s, g)$ that generalizes across both states and goals. By sharing representations across goals, UVFAs enable efficient transfer learning and zero-shot generalization to new objectives. Schaul et al. (2015) demonstrated UVFAs on several Atari games including Ms. Pac-Man, where goal conditioning on pellet collection improved learning efficiency.

The key advantage of GCRL is its ability to learn from off-policy experience that was generated with different goals in mind, enabling more sample-efficient learning in multi-objective environments.

2.3. Hindsight Experience Replay

Hindsight Experience Replay (HER) (Andrychowicz et al., 2017) is a data augmentation technique that enables learning from failed episodes by relabeling them with achieved goals. The core insight is that even when an agent fails to reach its intended goal, the trajectory contains valuable information about reaching the goal it actually achieved.

The HER algorithm operates as follows: After executing an episode with goal g, sample k additional goals from the episode (typically from future states using the "future" strategy) and store augmented transitions with these relabeled goals. For each transition $(s_t, a_t, r_t, s_{t+1}, g)$, HER creates additional transitions $(s_t, a_t, r'_t, s_{t+1}, g')$ where g' is an achieved goal from a future timestep and r'_t is recomputed according to the relabeled goal.

Originally developed for robotic manipulation tasks with

sparse binary rewards, HER dramatically improves sample efficiency by converting every failed episode into multiple successful training examples. This is particularly valuable when success is rare during early training.

2.4. Related Work

Recent work has explored various aspects of GCRL in different domains. Bortkiewicz et al. (2025) developed infrastructure for accelerating GCRL research across multiple benchmarks. Park et al. (2025) proposed transitive reinforcement learning as an alternative approach to compositional goal achievement. In the Atari domain, Verma et al. (2025) investigated transfer learning across games but did not specifically address goal-conditioned learning.

To our knowledge, no prior work has systematically applied HER to Atari Seaquest. While Schaul et al. (2015) demonstrated GCRL on other Atari games, Seaquest's unique combination of multiple competing objectives (diver rescue, oxygen management, enemy avoidance) and historically poor DQN performance makes it a distinct challenge for goal-conditioned approaches.

3. Approach

3.1. Environment

We conduct all experiments in the `ALE/Seaquest-v5` environment provided by the Arcade Learning Environment (ALE) (Bellemare et al., 2013).

Here, each observation consists of the RAM state from the Atari console (128 bytes) at a single time step (frame), plus the number of resurfaces as an additional byte. The RAM state contains game variables such as oxygen level, position of the submarine, diver counts, number of lives remaining, etc. The action space consists of all 18 legal actions on the Atari console. Although we incorporate reward shaping for our proposed algorithm, the conventional reward is represented by the in-game score, which we use to compare the performances of different agents. Finally, the episode terminates once the player has lost all of their lives (either by making contact with enemies or running out of oxygen).

3.2. Goal Representation

We define our goals in the form (N, M), where N is the number of divers collected, and M is the number of times resurfaced.

To account for differences in scale between the two dimensions, we then apply element-wise min-max normalization to (N, M), with the minimum at $(0, 0)$ and the maximum at $(N_{\text{max}}, M_{\text{max}}) = (6, 6)$. We set $N_{\text{max}} = 6$ because the player can only hold up to 6 divers between resurfaces, and $M_{\text{max}} = 6$ because the player must resurface with at least one diver (otherwise losing a life), leading to a maximum of 6 resurfaces.

Initially, we had experimented with goal representations of *three* dimensions (instead of two) but found that they were excessively complex. For more information, see Initial Experiments.

3.3. Deep Q-Learning

We build upon the deep Q-network (DQN) framework of Mnih et al. (2013), using an experience replay buffer and a target network that is periodically updated (every 10^4 steps). However, our approach differs in three key aspects.

First, to address potential overestimation of Q-values, we employ the *double DQN* (DDQN) algorithm proposed by van Hasselt et al. (2015). In particular, van Hasselt et al. (2015) demonstrated that DDQN can improve performance on Seaquest and other Atari games when properly tuned.

Second, because our target is now a *goal-conditioned* Q-value, the input to both the online and target networks consists of the *concatenation* of the observation and the normalized goal vector.

Finally, because the observation consists of the RAM state plus the number of resurfaces (129 bytes in total), we can employ a multi-layer perceptron (MLP) architecture with three hidden layers (size $512 \times 256 \times 128$). This differs from the setup in Mnih et al. (2013), in which the inputs are pixel images of the game, motivating a convolutional architecture.

3.4. Hindsight Experience Replay

To enhance our DDQN model, we employ the Hindsight Experience Replay (HER) method from Andrychowicz et al. (2017). Specifically, for each transition within a given episode that did not achieve the desired goals, we sample $k = 4$ future transitions and use their achieved goals as relabeled targets. This creates four new transitions that are relabeled as "successful," which we add to the DQN replay buffer alongside the original transition. These relabeled transitions encourage the model to learn from episodes in which it did not achieve the desired goal.

3.5. Curriculum Learning

The ultimate goal in our curriculum is $(N, M) = (6, 1)$, since collecting the maximum of 6 divers and then resurfacing results in a 1000-point bonus in the game (after which the agent may attempt repeat the same goal). However, it is difficult for the agent to directly learn from this goal, so we incorporate a two-phase curriculum for the agent to gradually progress towards this goal:

1. Phase 1: In the *Collection* phase, the agent focuses on learning only to collect divers. In particular, the desired goal is initially set to $(N, M) = (1, 0)$. Then, for $n = 1, 2, \ldots, 6$, once the agent achieves the desired goal $(n, 0)$ with success rate α_n over the last $L = 150$ episodes, the goal is then advanced to $(n+1, 0)$, except in the case of $n = 6$, in which case the agent advances to the Resurface phase.

2. Phase 2: In the *Resurface* phase, the agent repeats the Collection phase except with a resurface at the end (i.e., $M = 1$). Thus, the agent progressively learns to achieve the goals $(N, M) = (1, 1), (2, 1), \ldots, (6, 1)$.

To allow for more exploration immediately following advancement, we increase ϵ (for our ϵ-greedy policy) after each advancement and decay it subsequently.

3.6. Reward Shaping

HER is particularly effective with sparse binary rewards (Andrychowicz et al., 2017), making it useful in the standard GCRL setup where the (goal-conditioned) reward is defined as 0 if the goal is achieved and -1 otherwise (Liu et al., 2022). However, initial experiments suggested that this reward structure was insufficient for Seaquest's complex state space and our multi-step curriculum learning setup (see Curriculum Learning and Initial Experiments). Thus, we employ a custom reward function with the following components (Figure 2):

- Goal completion: $+50$ if the desired goal is achieved
- Survival: $+20$ for each remaining life, -10 for each life lost
- Diver bonuses: $+5$ for each diver collected
- Oxygen: -5 if oxygen level is below threshold $\beta = 16$

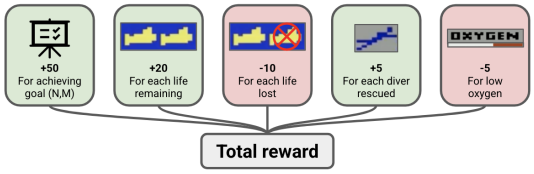

Figure 2. A visual representation of the reward structure we employed.

3.7. Prioritized Experience Replay

Due to the high occurrence of failures and subsequent HER relabeling, the replay buffer tends to become populated with transitions with easier achieved goals (e.g., $(N, M) = (1, 0)$) instead of those with more difficult achieved goals (e.g., $(N, M) = (3, 1)$), which provide deeper learning opportunities. As a result, we implement *priority sampling* to give additional sampling weight to transitions with sufficiently many divers, sufficiently high reward, or a successful resurface (for Phase 2).

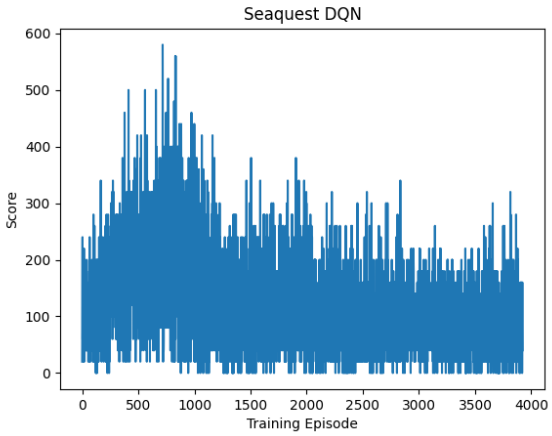

Figure 3. Rewards observed as a result of catastrophic forgetting

Yet, our experiments suggest that complete priority sampling can lead to *catastrophic forgetting*, where the agent forgets how to collect even one diver over the training iterations (Figure 3). As a result, we employ a *mixed sampling* scheme in which priority sampling is used for 60% of the batch, and uniform sampling for the remainder.

3.8. Implementation Details

Unless otherwise specified, we used a replay buffer size of $2,000,000$ transitions, a batch size of 256, a learning rate of 10^{-4}, and a discount rate of $\gamma = 0.99$.

4. Experimental Results

4.1. Experimental Setup

To evaluate the effectiveness of goal-conditioned DQN+HER, we trained DQN agents under several different configurations. As a baseline, we used the vanilla DQN with the standard reward of in-game score.

We trained each configuration for 10^7 steps and evaluated it by assessing episodic in-game scores over all training episodes. Another key point of comparison is the progress that each agent made in curriculum learning, i.e., the last value of (N, M) it was able to progress to. We also evaluated agent performance qualitatively by rendering episodes (i.e., watching the agent play, visually).

The training runs were performed on NVIDIA T4 GPUs

(through Google Cloud/Colab), as well as local CPUs.

4.2. Initial Experiments

Before moving forward with the approach described in Section 3, we tested the following alternative approaches, each of which had its own drawbacks. These drawbacks motivated us to change various aspects of the DQN+HER approach into those described in Section 3.

Sparse binary rewards. In GCRL, it is natural to define the goal-conditioned reward to be 0 if the goal is achieved and -1 otherwise (Liu et al., 2022). One can then apply HER, leveraging its effectiveness with sparse binary rewards (Andrychowicz et al., 2017). However, when we tried this, our replay buffer became filled with failures (-1), especially as the goals became more difficult. In particular, the agent simply did not receive enough positive feedback to learn from its mistakes, and so we decided to pursue a more complex reward strategy through reward shaping (Section 3.6).

Three-dimensional goals. We initially defined goals in the form (N_a, N, M), which represent the number of attackers killed, the number of divers rescued, and the number of resurfaces, respectively. However, we deemed this to be excessively complex – in particular, the number of attackers N_a should be implicitly learned instead of explicitly specified as a goal, so we instead switched to a two-dimensional goal structure (N, M).

Oxygen as a goal. We attempted to add oxygen level as an additional goal dimension to discourage the agent from letting its oxygen deplete. However, this led to a reward hacking situation in which the agent tended to simply stay on the surface and avoid diving down in order to maintain full oxygen. This suggests that the need to avoid oxygen depletion is better incorporated elsewhere in the model (e.g., in the reward function, which we did) instead of as an explicit goal.

4.3. Main Results

In general, the average episodic performance of DQN+HER did not exceed that of vanilla DQN (Figure 4) – both plateaued around 200-250 points. Moreover, with regard to the learning curriculum, DQN+HER was unable to consistently rescue $N = 4$ divers with $M = 0$ resurfaces. However, there are about 30 episodes in which DQN+HER achieved an outlier score of > 1000. This is because the player receives a 1000-point bonus for completely filling its capacity of 6 divers and resurfacing with all of them, which the agent here was able to achieve this on rare occasions. This provides evidence of goal-directed learning and suggests that DQN+HER has a higher score ceiling than vanilla DQN.

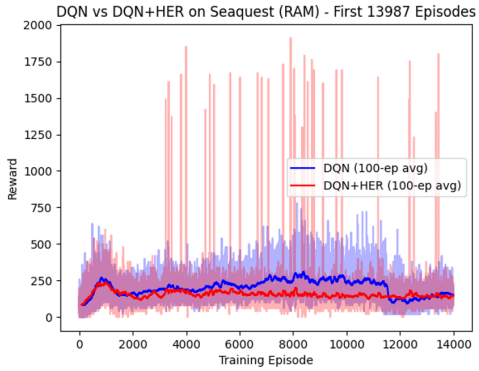

Figure 4. Episodic rewards (light) over training episodes for DQN (blue) and DQN+HER (red), with 100-episode running averages (dark) for each. The reward here is the *in-game score*, not the goal-conditioned one used in training.

Qualitatively, the poor performance of DQN+HER can likely be attributed to reward misspecification. In particular, renderings of the episodes suggest that the agent tended to stay in a single location and shoot blindly, scoring points for staying alive but not making significant progress towards rescuing divers and resurfacing.

We attempted to improve performance by making some architectural modifications to the DQN+HER setup. First, we tried adding layer normalization and dropout layers to the neural networks in order to improve training stability and regularization. We also attempted to expand the input to include RAM states for the last *four* frames (instead of just the most recent frame) in order to learn the dynamics model of the environment (e.g., moving enemies). The resulting agent was unable to collect more than $N = 3$ divers, so it performed worse than the single-frame scenario.

4.4. Ablation Study: Hyperparameter Tuning

To assess the impact of hyperparameter settings, we retrained the DQN+HER agent with the following alternative settings: (i) batch size of 512 instead of 256 and (ii) replay buffer size of 4 million instead of 2 million.

We found that the increase in batch size did not lead to better agent performance – the agent was still only able to reach $N = 4$ divers rescued. On the other hand, the increase in replay buffer size offered marginal improvement – the agent was able to reach $N = 5$ divers rescued.

4.5. Ablation Study: Soft Actor Critic

To investigate whether the policy learning algorithm impacts curriculum progression, we replaced Double DQN with Soft

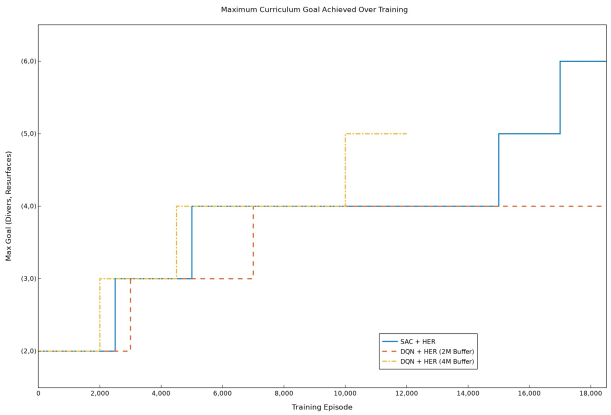

Figure 5. Comparison of the number of iterations taken to progress in the curriculum for DQN+HER with replay buffer size 2 million (red, dashed) vs. SAC+HER (blue, solid). vs. DQN + HER with replay buffer size of 4 million

Figure 6. Average episodic scores (top) and HER rewards (bottom) obtained by the SAC+HER agent.

Actor-Critic (SAC) (Haarnoja et al., 2018) while retaining the HER replay buffer, curriculum structure, and reward shaping from Section 3.

SAC optimizes a stochastic policy by maximizing both expected reward and policy entropy:

$$J(\pi) = \mathbb{E}_{\tau\sim\pi}\left[\sum_{t=0}^{\infty}\gamma^t\Big(R(s_t,a_t) + \alpha\mathcal{H}\big(\pi(\cdot|s_t)\big)\Big)\right] \quad (3)$$

where α controls the entropy regularization strength. This entropy bonus encourages broader exploration compared to epsilon-greedy strategies.

We trained both the DQN+HER and the SAC+HER models for 20 million training steps. Like before, the DQN model was only able to progress to $N = 4$ divers with $M = 0$ resurfaces. However, the SAC model managed to collect the maximum $N = 6$ divers and was learning to master resurfacing by the end of training (Figure 5). Qualitatively, the SAC agent survived longer as it collected more divers, suggesting that it had learned how to move around, avoid enemies, and collect divers, to a greater degree than the DQN agent did. This is supported by Figure 6, which shows that the average episodic reward improved as training progressed, unlike in DQN (where the average tended to plateau).

These results suggest that exploration strategy can significantly impact curriculum learning effectiveness in goal-conditioned settings. The entropy-regularized exploration in SAC appears better suited for discovering strategies required at higher curriculum levels than the epsilon-greedy exploration in DQN.

5. Conclusion

We investigated the application of goal-conditioned reinforcement learning with Hindsight Experience Replay to Atari Seaquest, a game characterized by multiple competing objectives and historically poor DQN performance.

Key findings. Our experiments reveal that HER enables goal-directed learning in Seaquest, as evidenced by occasional high-reward episodes demonstrating successful goal completion sequences. However, the combination of HER with Double DQN did not consistently outperform baseline DQN, plateauing at moderate curriculum levels. Replacing DQN with Soft Actor-Critic substantially improved curriculum progression, suggesting that exploration strategy is critical for goal-conditioned learning effectiveness.

Lessons learned. Insights that emerged from our work include:

- **Dense reward shaping is necessary.** Seaquest's complex 128-byte RAM state space proved incompatible with sparse binary rewards, requiring custom reward shaping for any learning to occur.
- **Exploration strategy matters.** The contrast between DQN+HER (plateaued at 4 divers) and SAC+HER (reached 6 divers) highlights that goal relabeling alone is insufficient without effective exploration.
- **Increased buffer size can promote learning.** Doubling the size of the experience replay buffer from 2 million to 4 million allowed the agent to progress from rescuing $N = 4$ divers to $N = 5$ divers. In general,

more hyperparameter tuning (while computationally intensive) can lead to improved performance.

Limitations. Our work faces several constraints. Training on CPU and basic GPU hardware limited comprehensive hyperparameter exploration and prevented multiple random seed evaluation. The SAC+HER results, while promising, represent a single training run. Additionally, our dense reward shaping approach departs from HER's original sparse-reward design, potentially limiting insights about HER's core mechanism.

Future directions. Several promising research directions emerge:

- **Alternative goal representations.** Learned goal embeddings or state-space goals may better capture Seaquest's objectives than our hand-crafted (N, M) representation.
- **Other GCRL methods.** Goal-conditioned supervised learning or other GCRL approaches beyond HER may prove more effective.
- **Transfer to related domains.** Testing our approach on other multi-objective Atari games (e.g., Montezuma's Revenge) could reveal generalizable insights about GCRL in discrete-action environments.
- **Improved reward shaping.** Better reward designs that encourage enemy avoidance while maintaining goal structure could address the termination bottleneck we observed.

Our work highlights challenges in adapting HER from sparse-reward robotics tasks to complex discrete-action games, particularly the importance of exploration strategy and reward shaping.

6. Contributions

Sanjeev Anthia Ganesh contributed to the core algorithm design and implementation for DQN, HER, and SAC.

Kevin Liu contributed to experimental design and execution, as well as development of all project deliverables.

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A. Appendix

The codebase can be found at https://github.com/Sanjeev-AG/cs234_final_project.